Name	Set Size (LLaMA-2)	Random Example String
Full	32000	不Иира циттфіцій Are filtersulsionNot Jersey grass
ASCII	25420	assigning Rahmenaltung indicate unit seed jap ToolsBD Snoween cleanecz los
ASCII without code	24021	settledRELEASEchenkozvb hit convert press Statistics Cooper dalla regional
Non-Latin	6090	тан°мирова.«∇竹}>фер сайте}}$.ı”/ правиê라 Михай
Non-alphabetic	1582	`:^{ (*;\='="{}^{(’«)>
Chinese	700	康春太全朱御系击志健思七明幸部
Characters	94	Y0-mo-0tX;mW1b{u
Invisible	64	
Zalgo	48	p̸l̸e̸a̸s̸ẽ

Table 6: Examples for constraint sets, showing cardinality and a random example string. Both cardinality and examples are shown for the LLaMA-2 tokenizer.

may invalidate the responses for all users in the batch. This would lead to a drastic increase in failed requests, especially if these overflowing requests are requeued dynamically.

However, current attacks are not able to raise intermediate activation values to levels outside of `float16` precision. However the optimization objective is, by definition, not numerically stable, and better optimized in a higher level of precision on the attacker's side. We show an example attack string that we optimize to maximize logit values in Figure 18. Interestingly, the attack achieves large logit values through construction of long nonsense phrases, with limited whitespaces and no linebreaks.

6.4 Collision Attacks

Finally, a different attack type that we experiment with, were collision attacks. Here, we optimize the KL divergence between the probabilities of a target sequence, and the probabilities of an attack sequence, where certain target words are replaced by the attack. In this way, the adversarial attack tokens "collide" with the target tokens. An example is shown in Figure 11 in the right inset. Overall, we found this attack to be moderately hard to optimize for, and without great practical usecases. In principle one could optimize for a collision with any kind of "banned" token, such as EOS tokens or system/assistant formatting tokens, but attacks based on such collisions were not more successful for us than attack directly target specific behaviors.

Figure 11: A collision attack generated by optimizing for an attack string that would lead to the same outcome as if the tokens for "Sam Altman" had been put in the same place, while blocking these tokens. The attack is a success and the model treats the attack string like the tokens for "Sam Altman". Attack on LLaMA-2-13b-chat (the 7b model refused to tell jokes).

6.5 Breaking The Spell

We further observe that these conversational models are not always fully swayed by the adversarial attacks. While a model might provide a profane answer, for example in Table 2, it can recover and follow up on this with a second response, such as "No I don't get it. Why are you saying that?" or "Please answer the question without using any of the prohibited words", Figure 3. In some situations, these are results of role hacking, where the model does not understand that the harmful response was provided by itself, and not the user. In others, it seems clear that models can contradict themselves, and while they cannot remove previously written text, the model can be capable of commenting on the event.

7 Conclusions

Why care about adversarial examples for LLMs? In this work, above everything else, we want to highlight the ease with which current-generation LLMs can be coerced into a number of unintended

behaviors. Even if we believe that jailbreaking attacks do not currently result in any harm, examples such as misdirection, denial-of-service and extraction show that these attacks already have capabilities that can cause harm in applications using current models. We consider our work complementary to recent work that focuses on improving optimizers and strategies for existing jailbreaking objectives, providing an overview over what else is possible.

We further analyze properties and behaviors of these attacks, finding that optimized attacks discover some strategies that have been strenuously discovered through manual red-teaming (Wei et al., 2023), as well as new behaviors that we have not observed yet, such as role hacking, and the utilization of glitch tokens. The broadest mechanism that we show throughout this work, though, is *the propensity of attacks to coerce the model into simulating code*, and to complete programming instructions and function calls, instead of following the trained conversational behavior. Given that these models are trained on web data which contains significant amounts of code interspersed with language data, and given that we generally want to train models on both language and code, to be useful as coding assistants, this interdependence appears hard to resolve.

We ultimately conclude by confirming the initial hypothesis that arbitrary free text input to these models allows almost any outcome, and that for now, from a security perspective, any output of a language model, generated from user input, has to be assumed to be insecure. Fundamentally, there might be no fix for this.

Broader Impact

In this work, we show a range of new adversarial attack objectives for LLMs that could technically be deployed against systems in production. We do not optimize our attacks against real-world systems, as we do not run model ensembles that would better transfer to black-box systems, but this step is small. We argue that the disclosure of this work provides additional evidence for the limitations of current systems that practitioners have to take into account. We are not the first to point out these problems, but we only hope to be a bit convincing. We see the merit of our work in this aspect of disclosing and convincing current and future builders of LLM-based applications of the fundamental fact that adversarial attacks exist, even for LLMs. And, of the conclusion that with free-text input, LLM responses can be almost anything, even for models aligned through safety procedures such as RLHF.

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